Superpixel-Based Autoencoder-Like Nonnegative Tensor Factorization for Hyperspectral Unmixing

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***Abstract*—Hyperspectral unmixing is significant for advancing remote sensing (RS) applications, aiming at extracting the spectra of pure materials (called endmembers) and obtaining their proportions (called fractional abundances) from an observed hyperspectral image (HSI). Nonnegative matrix factorization (NMF) is a popular technique for hyperspectral unmixing; however, it does not effectively preserve the spatial and spectral correlation of HSIs and fully characterize spectral variability. To overcome these limitations, we propose a novel superpixel-based autoencoder-like nonnegative tensor factorization (SANTF) model for hyperspectral unmixing. Specifically, drawing inspiration from the architecture of autoencoder, an autoencoder-like nonnegative tensor factorization (ANTF) model is constructed to directly project the hyperspectral data into abundance space. To further exploit the local spatial information, the superpixel strategy is incorporated into the ANTF framework, thereby building the SANTF model. Note that the superpixel cubes are jointly factorized instead of being represented as an average. Meanwhile, each superpixel cube generates one endmember matrix to capture spectral variability. Subsequently, a robust double weighted endmember (rDWE) constraint is designed to obtain the consensus endmember matrix adaptively. Moreover, the tensor nuclear norm (TNN) constraint is employed to enhance the low-rank characteristic of abundance tensor. The experimental results on both synthetic and real HSI datasets, compared with several unmixing methods, demonstrate that the proposed SANTF method can achieve superior unmixing performance.**

***Index Terms*—Autoencoder, endmember constraint, hyperspectral unmixing, nonnegative tensor factorization (NTF), superpixel.**

I. Introduction

HYPERSPECTRAL images (HSIs) simultaneously possess rich spectral and spatial information, enabling precise identification and analysis of ground covers [1], [2]. However, caused by the limitation of spatial resolution of the spectrometer and complexity of materials, a single pixel usually contains multiple components, a phenomenon known as mixed pixel. Hyperspectral unmixing aims to accurately extract spectra of pure materials (i.e., endmembers) and their corresponding proportions (i.e., abundances) from mixed pixels, which is of great significance in the field of remote sensing (RS) images [3].

Spectral mixture models encompass both linear mixture model (LMM) and nonlinear mixture model (NLMM), categorized according to the mixture scale [4], [5]. LMM considers a checkerboard-type macroscopic mixture, where an observed pixel is derived from a linear combination of the endmembers, weighted by their associated abundances [6]. Most unmixing methods are proposed under the LMM due to its advantages of simplicity and efficiency. NLMM involves a microscopic mixture, allowing to characterize the interactions at intimate [7] and multilayered levels [8], [9], [10].

Based on spectral mixture models, many unmixing algorithms have been developed in recent years. Four main classes of techniques are geometrical, sparse regression, statistical, and deep learning (DL)-based techniques [1]. Geometrical techniques only focus on obtaining endmembers that correspond to the vertices of the simplex confined to the mixed pixels, including pure pixel-based methods [11], [12] and minimum volume (MV)-based methods [13], [14]. Sparse regression-based techniques estimate abundances using an available spectral library [15], [16], [17]. However, their performance may be limited by the high spectral coherence present in the library and diverse imaging conditions. Statistical techniques investigate the properties of HSIs and address the unmixing problem from an unsupervised perspective [18]. Particularly, nonnegative matrix factorization (NMF) is a representative model for hyperspectral unmixing. DL-based techniques have been extensively studied to tackle mixed pixels due to their powerful abilities of representation [19]. Typical architectures are autoencoder [20], [21], [22] and convolutional neural networks (CNNs) [23], [24]. Nonetheless, these methods generally suffer from a lack of physical interpretability and necessitate careful tuning of network nodes to achieve satisfactory unmixing performance [25].

NMF has garnered significant attention due to its capability to simultaneously extract endmembers and their associated abundances. For instance, He et al. [26] integrated a weighted

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sparse regularizer and a total variation (TV) regularizer into NMF to enhance the sparsity and piecewise smooth of the abundance matrix. In [27], a spectral–spatial robust NMF (SSRNMF) model is presented to achieve robustness against band noise and pixel noise simultaneously. By combining two types of self-similarity, an NMF-based method is proposed to explore both global spatial information and local spectral information [28]. In addition, a higher order graph regularizer is constructed to better preserve the global spatial structure of the data in [29]. Although NMF-based approaches demonstrate competitive unmixing performance, they generally destroy the internal structure of hyperspectral data due to the requirement for unfolding processing.

To overcome the limitations of the NMF model, nonnegative tensor factorization (NTF) has been introduced to preserve the intrinsic structure information of HSIs [30]. As a high-dimensional extension of a matrix, tensors are more suitable for representing HSIs. The reported NTF-based unmixing algorithms have three development directions [31]. First, a matrix-vector NTF (MV-NTF) method has been proposed by decomposing the HSI tensor into a sum of several component tensors [32]. To make full use of local spatial information, a TV regularizer is further incorporated into MV-NTF to generate smooth abundance maps [33]. Moreover, an endmember independence constraint is leveraged to avoid obtaining similar endmember estimations [34]. Second, by adopting the definition of tensor mode-n product, many NTF-based methods have been developed to explore more characteristics, such as spatial similarity [35], nonlocal similarities [36], [37], and nonlinear interaction [31], [38]. Finally, to provide flexibility regarding the rank of the associated tensor, a low-rank regularization is employed to constrain the 4-D endmember tensor that characterizes spectral variability [39] or the estimated abundance tensor [40]. Nevertheless, existing NTF methods only reconstruct the original data from the endmembers in abundance space, i.e., decoder. Although Guan et al. [41] have constructed an autoencoder-like nonnegative tensor decomposition (ANTD), it leverages Tucker decomposition for community detection. As such, we adopt the definition of the tensor mode-3 product to provide interpretability for the hyperspectral unmixing problem. Furthermore, local spatial information and spectral variability have not been fully explored, leading to an inadequate ability to depict spectral similarity and abundance dependence among adjacent pixels.

In this article, we propose a superpixel-based autoencoder-like NTF (SANTF) method to address the problem of hyperspectral unmixing, which considers spectral variability at the superpixel level and combines spatial information. Specifically, superpixels are generated and processed into regular cubes to exploit the local spatial information. Then, the resulting superpixel cubes are jointly decomposed using the autoencoder-like NTF (ANTF), which contains two parts: NTF encoder and NTF decoder. NTF encoder directly projects the original data space into the abundance space, while the decoder concentrates on reconstructing the original data using endmembers and abundances. Owing to each superpixel generating one endmember matrix, spectral variability can be characterized from a superpixel perspective. Our main contributions can be summarized as follows.

1) An ANTF is first exploited to address the mixed pixels, which integrates both the NTF decoder and NTF encoder together. The innovative framework not only avoids structural information loss caused by the HSI cube unfolding processing but also can directly project the original data into abundance space and effectively reconstruct the original data.
2) Considering the local spatial information and spectral variability, the superpixel strategy is applied to achieve shape-adaptive homogeneous regions to construct the SANTF model. Particularly, superpixel cubes are directly input to preserve details instead of being represented as an average. In this case, each superpixel cube yields one endmember matrix to capture spectral variability at the superpixel level.
3) A robust double weighted endmember (rDWE) constraint is designed from internal and external superpixel perspectives to learn the high-quality consensus endmember matrix adaptively. Meanwhile, the tensor nuclear norm (TNN) constraint is employed to enhance the low-rank characteristic of abundance tensor.
4) Finally, we derive the optimization method by alternating direction method of multipliers (ADMM) aimed at efficiently solving the objective function of SANTF. Experiments are conducted to testify the validity of the proposed SANTF method on the synthetic and different real hyperspectral datasets.

The organization of this article is structured as follows. In Section II, we provide a concise overview of tensor concepts and operations, as well as LMM and NTF. Section III presents the proposed SANTF algorithm, detailing its theoretical foundations and implementation issues. Experiments are performed and the resulting findings are discussed in Section IV. Finally, Section V concludes this article by summarizing the key insights and implications of our research.

II. Related Work

A. Concepts and Operations

Definition 1 (The Mode-n Matricization): For an Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$, the mode-$n$ matricization organizes the mode-n vectors into columns of the resulting matrix, represented as $\mathbf{X}_{(n)} \in \mathbb{R}^{I_n \times (I_1 \cdots I_{n-1} I_{n+1} \cdots I_N)}$.

Definition 2 (The Mode-n Product): The mode-n product of a tensor $\mathcal{X}$ by a matrix $\mathbf{U} \in \mathbb{R}^{J \times I_n}$ is a tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$, denoted by

$$\mathcal{Y} = \mathcal{X} \times_n \mathbf{U} \tag{1}$$

where the components of tensor $\mathcal{Y}$ are computed by $\mathcal{Y}(i_1, \ldots, i_{n-1}, j, i_{n+1}, \ldots, i_N) = \sum_{i_n=1}^{I_n} \mathcal{X}(i_1, \ldots, i_N)\, \mathbf{U}(j, i_n)$. The formulation using unfolded tensors is expressed as

$$\mathbf{Y}_{(n)} = \mathbf{U}\mathbf{X}_{(n)}. \tag{2}$$

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Definition 3 (The Norm of a Tensor): The Frobenius norm of the tensor $\mathcal{X}$ is defined as

$$\|\mathcal{X}\|_F = \sqrt{\sum_{i_1=1}^{I_1}\sum_{i_2=1}^{I_2}\cdots\sum_{i_N=1}^{I_N}\mathcal{X}_{i_1 i_2 \ldots i_N}^2}. \tag{3}$$

Definition 4 (Tensor Nuclear Norm): The TNN of $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}$ is defined as

$$\|\mathcal{X}\|_{\text{TNN}} = \sum_{i=1}^{I_3}\sum_{j}\sigma_j\left(\mathbf{X}^{(i)}\right) \tag{4}$$

where $\sigma_j(\mathbf{X}^{(i)})$ denotes the jth singular value of $\mathbf{X}^{(i)}$.

B. Linear Mixture Model

In the LMM, an observed pixel spectrum is assumed to be linearly combined by endmembers along with their abundance proportions. In order to preserve the intrinsic 3-D spatial information, the HSI data $\mathcal{X} \in \mathbb{R}^{r\times c\times B}$, including r rows, c columns, and B bands, are directly factorized into one nonnegative matrix $\mathbf{A}$ and one nonnegative three-order tensor $\mathcal{S}$, given as

$$\mathcal{X} = \mathcal{S}\times_3 \mathbf{A} + \mathcal{N} \tag{5}$$

where $\mathbf{A} \in \mathbb{R}^{B\times M}$ refers to the spectra with M endmembers, $\mathcal{S} \in \mathbb{R}^{r\times c\times M}$ is the corresponding abundance tensor, and $\mathcal{N} \in \mathbb{R}^{r\times c\times B}$ represents the Gaussian noise. $\mathcal{S}$ is restricted by abundance nonnegative constraint (ANC) (i.e., $S_{ijk} \geq 0$) and abundance sum-to-one constraint (ASC) (i.e., $\mathcal{S}^g \times_3 \mathbf{1}_M^T = \mathcal{I}$), here $i = 1, 2, \ldots, r$, $j = 1, 2, \ldots, c$, $k = 1, 2, \ldots, M$, $\mathbf{1}_M \in \mathbb{R}^{M\times 1}$ is a column vector of all ones, and $\mathcal{I} \in \mathbb{R}^{r\times c\times 1}$ is a tensor of all ones.

C. NTF and Its Variants

Based on LMM, NTF-based methods using the definition of the tensor mode-n product can be summarized as follows:

$$\min_{\mathbf{A},\mathcal{S}} \frac{1}{2}\|\mathcal{X} - \mathcal{S}\times_3 \mathbf{A}\|_F^2 + \varphi(\mathcal{S}) \tag{6}$$

where $\mathcal{S} \in \mathbb{R}^{r\times c\times M}$ is the abundance tensor; $\varphi(\mathcal{S})$ is a regularization term for abundances, e.g., a weighted TV regularizer [35]; and a multimode low-rank learning regularization and sparsity constraint [37].

Furthermore, given a spectral library, NTF-based methods developed for sparse unmixing can be formed as

$$\min_{\mathcal{S}} \frac{1}{2}\|\mathcal{X} - \mathcal{S}\times_3 \mathbf{A}\|_F^2 + \psi(\mathcal{S}) \tag{7}$$

where $\mathbf{A}$ is the known spectral library and $\psi(\mathcal{S})$ is a regularization term for abundances such as the nonlocal low-rank and TV regularizations [36] and three constraints (i.e., TV, collaborative sparsity, and nonlocal tensor low-rank) [42]. Similarly, the tensor mode-2 product is leveraged since mode-2 of the third-order tensor is the spectral mode [43].

III. Methodology

A. Proposed SANTF Model

In general, (6) and (7) define the reconstruction error, which is regarded as a decoder part in the architecture of an autoencoder. However, it is essential to incorporate an encoder part that can directly map the HSI data into the abundance space using endmember representation. Specifically, $\mathcal{S}$ can be expressed through the mode-3 product of $\mathcal{X}$ and $\mathbf{A}$. However, in hyperspectral unmixing applications, it is difficult to accurately represent $\mathcal{S}$ by using $\mathcal{X}$ and $\mathbf{A}$ due to the nonorthogonality between the endmember spectra. To this end, an encoder weight matrix $\mathbf{E}$ is introduced to replace $\mathbf{A}$, resulting in $\mathcal{S} = \mathcal{X}\times_3 \mathbf{E}^T$. Consequently, by integrating both encoder and decoder components, an ANTF model is formed as

$$\begin{aligned}&\mathcal{C} = \frac{1}{2}\|\mathcal{X} - \mathcal{S}\times_3 \mathbf{A}\|_F^2 + \frac{\lambda}{2}\left\|\mathcal{S} - \mathcal{X}\times_3 \mathbf{E}^T\right\|_F^2\\ &\text{s.t. } \mathbf{A} \geq 0, \quad \mathcal{S} \geq 0,\ \mathcal{S}\times_3 \mathbf{1}_M^T = \mathcal{I}\end{aligned} \tag{8}$$

where λ is a balanced parameter and $\mathbf{E} \in \mathbb{R}^{B\times M}$ denotes the encoder weight matrix.

It is well known that the consideration of local spatial information significantly improves the unmixing performance of HSI. In contrast to the pixel patches extracted using the sliding window approach, superpixels can achieve better spatially homogeneous regions due to their adaptive shape and size. Consequently, the improved simple linear iterative clustering (SLIC) [44] is employed to generate G spatial groups characterized by similar spectral property and spatial adjacency. There are two parameters for improved SLIC: ω_s and ω, where ω_s balances the relative importance between the spatial and spectral similarities and ω controls the average size of the superpixels. A larger ω_s value prioritizes spatial similarity over spectral similarity, and a smaller ω_s value does the opposite. For ω, a smaller value may lead to inadequate exploitation of spatial information, while a larger value can compromise the preservation of spatial homogeneity. This process results in G different superpixels, denoted as $\mathcal{X}^g$. It is important to emphasize that these irregular local spatial groups must be refined into regular shapes by filling in zero. According to the ANTF model, each superpixel $\mathcal{X}^g$ can be factorized into one nonnegative matrix $\mathbf{A}^g$ and one nonnegative tensor $\mathcal{S}^g$ such that $\mathcal{X}^g \approx \mathcal{S}^g \times_3 \mathbf{A}^g$. Besides, in the encoder component, the abundance cubes of all superpixels are represented with the same encoder weight, expressed as $\mathcal{S}^g \approx \mathcal{X}^g \times_3 \mathbf{E}^T$. As such, a novel SANTF model is constructed, which is defined as

$$\begin{aligned}&\mathcal{C} = \sum_{g=1}^{G}\frac{1}{2}\|\mathcal{X}^g - \mathcal{S}^g\times_3 \mathbf{A}^g\|_F^2 + \frac{\lambda}{2}\left\|\mathcal{S}^g - \mathcal{X}^g\times_3 \mathbf{E}^T\right\|_F^2\\ &\text{s.t. } \mathbf{A}^g \geq 0, \quad \mathcal{S}^g \geq 0,\ \mathcal{S}^g\times_3 \mathbf{1}_M^T = \mathcal{I}^g\end{aligned} \tag{9}$$

where $g = 1, \ldots, G$, $\mathcal{X}^g \in \mathbb{R}^{r_g\times c_g\times B}$, $\mathbf{A}^g \in \mathbb{R}^{B\times M}$ is endmember matrix for the gth superpixel, $\mathcal{S}^g \in \mathbb{R}^{r_g\times c_g\times M}$ denotes the abundance cube for the gth superpixel, and $\mathcal{I}^g \in \mathbb{R}^{r_g\times c_g\times 1}$ is a tensor of all ones.

Each superpixel region exhibits homogeneity, indicating that its pixels are more likely to be composed of the same materials. Consequently, it is reasonable to assume that the endmember

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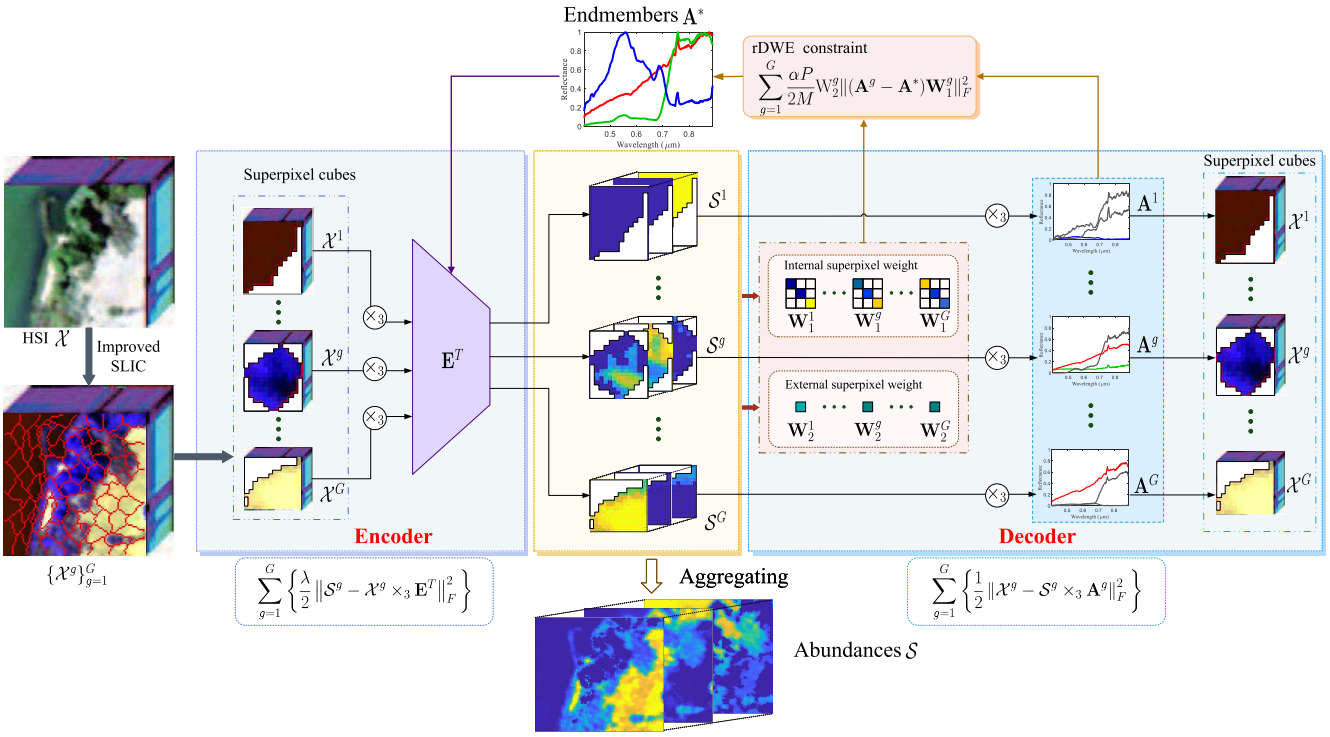


Fig. 1. Framework of the proposed SANTF model for hyperspectral unmixing.

spectra corresponding to the pixels within a superpixel are identical, i.e., the pixels within the gth superpixel region will yield one endmember matrix $\mathbf{A}^g$. As indicated by (9), this process results in G endmember matrices, collectively represented as $\{\mathbf{A}^g\}_{g=1}^G$. These G endmember matrices are all distinct, which can be utilized to characterize the spectral variability phenomenon presented in hyperspectral data. Specifically, as shown in Fig. 1, $\mathbf{A}^g$ derived from different superpixels might emphasize different spectral signatures. For instance, when the first superpixel $\mathcal{X}^1$ includes one signatures (i.e., $\mathbf{a}_3$). In this case, the corresponding signature $\mathbf{a}_3$ (blue line) within the resulting first endmember $\mathbf{A}^1$ can be learned effectively. However, estimating $\mathbf{a}_1$ and $\mathbf{a}_2$ (gray lines) is very difficult due to the absence of corresponding material information within $\mathcal{X}^1$. Similarly, if each pixel with another superpixel $\mathcal{X}^g$ consists of two signatures, i.e., $\mathbf{a}_1$ and $\mathbf{a}_2$, then these signatures can be learned effectively within the resulting endmember matrix $\mathbf{A}^g$. Moreover, if each pixel in the Gth superpixel $\mathcal{X}^G$ contains only one signatures, namely $\mathbf{a}_1$, in the resulting endmember $\mathbf{A}^G$, $\mathbf{a}_1$ can be learned effectively. It is noteworthy that there is both consensus and complementary information among the endmembers $\mathbf{a}_1$ derived from different superpixels like $\mathcal{X}^g$ and $\mathcal{X}^G$. Furthermore, the impact of spectral variability results in both consensus and complementary information among $\{\mathbf{A}^g\}_{g=1}^G$. Meanwhile, for the gth superpixel region, some small values may be obtained iteratively in the gth abundance cube, which could adversely affect the learning of the consensus endmember matrix. Thus, an rDWE constraint is proposed to alleviate the impact caused by these small values, guaranteeing high-quality consensus endmembers. Specifically, a robust operator is incorporated into weight $\mathbf{W}_1^g$, represented by the Cauchy loss $\varphi(s) = \ln(1 + s^2/\rho^2)$, where ρ is a scale parameter of the Cauchy distribution. Consequently, $\mathbf{W}_1^g$ is defined as

$$\mathbf{W}_1^g = \operatorname{diag}\left(\sum_{i,j}\varphi\left(\left[\mathcal{S}_{::1}^g\right]_{ij}\right),\ldots,\sum_{i,j}\varphi\left(\left[\mathcal{S}_{::M}^g\right]_{ij}\right)\right)\Big/\Phi^g \tag{10}$$

where $\Phi^g = \sum_{i,j,k}\varphi(\mathcal{S}_{ijk}^g)$ denotes a normalization constant, which is calculated in advance. According to (10), it will make the impact caused by the small values gradually approach zero, and thus, it is robust to small values in the abundance cubes. To this end, for learning the consensus endmember matrix $\mathbf{A}^*$, (9) is transformed into the following cost function:

$$\begin{aligned}\mathcal{C} = &\sum_{g=1}^{G}\frac{1}{2}\left\|\mathcal{X}^g - \mathcal{S}^g \times_3 \mathbf{A}^g\right\|_F^2 + \frac{\lambda}{2}\left\|\mathcal{S}^g - \mathcal{X}^g \times_3 \mathbf{E}^T\right\|_F^2 \\ &+ \frac{\alpha P}{2M}W_2^g\left\|\left(\mathbf{A}^g - \mathbf{A}^*\right)\mathbf{W}_1^g\right\|_F^2 \\ \text{s.t. } &\mathbf{A}^g \geq 0, \quad \mathbf{A}^* \geq 0,\ \mathcal{S}^g \geq 0, \mathcal{S}^g \times_3 \mathbf{1}_M^T = \mathcal{I}^g\end{aligned} \tag{11}$$

where α is a tunable regularized parameter and $W_2^g = N_g/P$, in which N_g denotes the total number of pixels in the gth superpixel and P represents the total number of all pixels. $\mathbf{W}_1^g$ is designed to capture the endmember spectra presumed to be present within the superpixel region itself, while W_2^g measures the influence of the number of pixels within the gth superpixel on the overall total number of pixels.

In addition, the abundance tensor generally exhibits a low-rank structure because of significant spatial correlation presented in the HSIs. To accurately represent the different ranks of each tensor slice, the TNN is utilized to encourage the low-rank characteristics of the abundance tensor. Thus,

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a novel SANTF model with rDWE and TNN constraints is proposed, whose cost function is defined as

$$\mathcal{C} = \sum_{g=1}^{G} \left\{ \frac{1}{2} \|\mathcal{X}^g - \mathcal{S}^g \times_3 \mathbf{A}^g\|_F^2 + \frac{\lambda}{2} \left\|\mathcal{S}^g - \mathcal{X}^g \times_3 \mathbf{E}^T\right\|_F^2 + \frac{\alpha P}{2M} W_2^g \left\|(\mathbf{A}^g - \mathbf{A}^*)\,\mathbf{W}_1^g\right\|_F^2 \right\} + \beta \|\mathcal{S}\|_{\text{TNN}}$$
$$\text{s.t. } \mathbf{A}^g \geq 0, \quad \mathbf{A}^* \geq 0, \ \mathcal{S}^g \geq 0, \ \mathcal{S}^g \times_3 \mathbf{1}_M^T = \mathcal{I}^g \tag{12}$$

where β is also a balanced parameter to control the tradeoff for each term. The framework of the proposed SANTF model is shown in Fig. 1.

B. Optimization

In order to optimize (12), an auxiliary tensor $\mathcal{H}$ requires to be introduced to easily optimize the TNN constraint under the subject to $\mathcal{S} = \mathcal{H}$. For the constraints $\mathcal{S} = \mathcal{H}$ and $\mathcal{S}^g \times_3 \mathbf{1}_M^T = \mathcal{I}$, by using the Lagrangian function, the augmented Lagrangian function of (12) is formulated as

$$\mathcal{C} = \sum_{g=1}^{G} \left\{ \frac{1}{2} \|\mathcal{X}^g - \mathcal{S}^g \times_3 \mathbf{A}^g\|_F^2 + \frac{\lambda}{2} \left\|\mathcal{S}^g - \mathcal{X}^g \times_3 \mathbf{E}^T\right\|_F^2 + \frac{\mu}{2} \left\|\mathcal{H}^g - \left(\mathcal{S}^g + \frac{\mathcal{D}^g}{\mu}\right)\right\|_F^2 + \frac{\delta}{2} \left\|\mathcal{I}^g - \mathcal{S}^g \times_3 \mathbf{1}_M^T\right\|_F^2 + \frac{\alpha P}{2M} W_2^g \left\|(\mathbf{A}^g - \mathbf{A}^*)\,\mathbf{W}_1^g\right\|_F^2 \right\} + \beta \|\mathcal{H}\|_{\text{TNN}}$$
$$\text{s.t. } \mathbf{A}^g \geq 0, \quad \mathbf{A}^* \geq 0, \ \mathcal{S}^g \geq 0 \tag{13}$$

in which μ and δ are penalty parameters. In order to optimize (13), a gradient descent method is utilized to alternately update one of the underlying variables while fixing the others at each iteration. As such, $\mathbf{A}^g$, $\mathcal{S}^g$, $\mathbf{A}^*$, $\mathbf{E}$, $\mathcal{H}$, and $\mathcal{D}$ are updated iteratively, whose update rules are developed in detail as follows.

1) Updating $\mathbf{A}^g$: For the gth superpixel, by simplifying (13) and transforming the tensor form into the matrix form, it can be seen that the update rule for $\mathbf{A}^g$ is given by minimizing the following cost function:

$$\mathbf{A}^g \leftarrow \arg\min_{\mathbf{A}^g} \frac{1}{2} \left\|\mathbf{X}_{(3)}^g - \mathbf{A}^g \mathbf{S}_{(3)}^g\right\|_F^2 + \frac{\alpha P}{2M} W_2^g \left\|(\mathbf{A}^g - \mathbf{A}^*)\,\mathbf{W}_1^g\right\|_F^2, \quad \text{s.t. } \mathbf{A}^g \geq 0 \tag{14}$$

where $\mathbf{X}_{(3)}^g$ denotes mode-3 unfoldings of tensor $\mathcal{X}^g$. Then, the derivative of (14) with respect to $\mathbf{A}^g$ is derived. By selecting an appropriate stepsize, the update rule for $\mathbf{A}^g$ can be given as

$$\mathbf{A}^g \leftarrow \mathbf{A}^g .* \left\{ \mathbf{X}_{(3)}^g \left(\mathbf{S}_{(3)}^g\right)^T + \frac{\alpha P}{M} W_2^g \mathbf{A}^* \mathbf{W}_1^g \left(\mathbf{W}_1^g\right)^T \right\} ./ \left\{ \mathbf{A}^g \mathbf{S}_{(3)}^g \left(\mathbf{S}_{(3)}^g\right)^T + \frac{\alpha P}{M} W_2^g \mathbf{A}^g \mathbf{W}_1^g \left(\mathbf{W}_1^g\right)^T \right\} \tag{15}$$

where $.*$ and $./$ denote the operations of product and division for each entry in the matrix, respectively.

2) Updating $\mathcal{S}^g$: Similarly, for the gth superpixel, by simplifying (13) and transforming the tensor form into the matrix form, it can be seen that the update rule for $\mathbf{S}_{(3)}^g$ is given by minimizing the following cost function:

$$\mathbf{S}_{(3)}^g \leftarrow \arg\min_{\mathbf{S}_{(3)}^g} \frac{1}{2} \left\|\mathbf{X}_{(3)}^g - \mathbf{A}^g \mathbf{S}_{(3)}^g\right\|_F^2 + \frac{\lambda}{2} \left\|\mathbf{S}_{(3)}^g - \mathbf{E}^T \mathbf{X}_{(3)}^g\right\|_F^2 + \frac{\mu}{2} \left\|\mathbf{H}_{(3)}^g - \left(\mathbf{S}_{(3)}^g + \frac{\mathbf{D}_{(3)}^g}{\mu}\right)\right\|_F^2 + \frac{\delta}{2} \left\|\mathbf{I}_{(3)}^g - \mathbf{1}_M^T \mathbf{S}_{(3)}^g\right\|_F^2 + \frac{\alpha P}{2M} W_2^g \left\|(\mathbf{A}^g - \mathbf{A}^*)\,\mathbf{W}_1^g\right\|_F^2. \tag{16}$$

In particular, the fifth term in (16) can be transformed into the following equation:

$$\mathcal{C}_{\mathbf{S}}^g = \frac{\alpha P \mathrm{W}_2^g}{2M\,(\Phi^g)^2} \sum_{m=1}^{M} \left\{ \left(\sum_p \varphi\left(\left[\mathbf{S}_{(3)}^g\right]_{mp} \right) \right)^2 \left\|\mathbf{a}_m^g - \mathbf{a}_m^*\right\|^2 \right\}. \tag{17}$$

As is well known, the derivation of $\varphi(s)$ with respect to s is expressed as $\partial\varphi(s)/\partial s = 2s/(\rho^2 + s^2)$. Therefore, the derivation of $\mathcal{C}_{\mathbf{S}}^g$ with respect to each matrix's element $[\mathbf{S}_{(3)}^g]_{mp}$ is given as

$$\frac{\partial \mathcal{C}_{\mathbf{S}}^g}{\partial \left[\mathbf{S}_{(3)}^g\right]_{mp}} = \frac{2\alpha P \mathrm{W}_2^g \left[\mathbf{S}_{(3)}^g\right]_{mp} \sum_i \varphi\left(\left[\mathbf{S}_{(3)}^g\right]_{mi}\right)}{M\,(\Phi^g)^2 \left(\left(\left[\mathbf{S}_{(3)}^g\right]_{mp} \right)^2 + \rho^2 \right)} \left\|\mathbf{a}_m^g - \mathbf{a}_m^*\right\|^2. \tag{18}$$

Then, the derivative of (16) with respect to $\mathbf{S}_{(3)}^g$ is derived. Subsequently, by adjusting $\mathbf{S}_{(3)}^g$ into the opposite direction of the gradient with an appropriate stepsize, the update rule for $\mathbf{S}_{(3)}^g$ is given as

$$\mathbf{S}_{(3)}^g \leftarrow \mathbf{S}_{(3)}^g .* \left\{ (\mathbf{A}^g)^T \mathbf{X}_{(3)}^g + \lambda \mathbf{E}^T \mathbf{X}_{(3)}^g + \mu \mathbf{H}_{(3)}^g + \delta \mathbf{1}_M \mathbf{I}_{(3)}^g \right\} ./ \left\{ (\mathbf{A}^g)^T \mathbf{A}^g \mathbf{S}_{(3)}^g + \mathbf{D}_{(3)}^g + (\lambda + \mu)\,\mathbf{S}_{(3)}^g + \delta \mathbf{1}_M \mathbf{1}_M^T \mathbf{S}_{(3)}^g + \frac{\partial \mathcal{C}_{\mathbf{S}}^g}{\partial \mathbf{S}_{(3)}^g} \right\}. \tag{19}$$

Furthermore, to obtain abundance cube $\mathcal{S}^g$ for the gth superpixel, we utilize the reshape operator, expressed as $\mathcal{S}^g = \text{reshape}_3\,(\mathbf{S}_{(3)}^g)$.

3) Updating $\mathbf{A}^*$: Similarly, by solving the derivative of (13) with respect to $\mathbf{A}^*$, the update rule for $\mathbf{A}^*$ is given as

$$\mathbf{A}^* \leftarrow \max\left\{ \left(\sum_{g=1}^{G} W_2^g \mathbf{A}^g \mathbf{W}_1^g \left(\mathbf{W}_1^g\right)^T \right) \left(\sum_{g=1}^{G} W_2^g \mathbf{W}_1^g \left(\mathbf{W}_1^g\right)^T \right)^{-1}, 0 \right\}. \tag{20}$$

4) Updating E: After obtaining the consensus endmember matrix $\mathbf{A}^*$, due to the implicit relationship between $\mathbf{E}$ and $\mathbf{A}^*$, $\mathbf{E}$ can be expressed directly by using $\mathbf{A}^*$. Therefore, the update rule for $\mathbf{A}^*$ is given as

$$\mathbf{E} \leftarrow \max\left\{ \mathbf{A}^* \left((\mathbf{A}^*)^T \mathbf{A}^* \right)^{-1}, 0 \right\}. \tag{21}$$

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5) Updating $\mathcal{H}$: After updating the cubes $\mathcal{S}^g$ from $g = 1$ to $g = G$ once, $\{\mathcal{S}^g\}_{g=1}^G$ require to be aggregated into abundance tensor $\mathcal{S}$. Then, by fixing other variables and omitting the items not related to $\mathcal{H}$, the subproblem of $\mathcal{H}$ is given as

$$\mathcal{H} \leftarrow \arg\min_{\mathcal{H}} \frac{\beta}{\mu} \|\mathcal{H}\|_{\text{TNN}} + \frac{1}{2} \left\| \mathcal{H} - \left(\mathcal{S} + \frac{\mathcal{D}}{\mu} \right) \right\|_F^2. \tag{22}$$

Then, $\mathcal{H}$ can be optimized according to [45].

Algorithm 1 Hyperspectral Unmixing via the SANTF Model

Input: HSI data $\mathcal{X}$, the number of endmembers M, and given parameters: ρ, μ, σ, δ, $\mu_{\max}$, $T_{\max}$, and ε; tunable parameters: α, β, and λ.

Initialize: $\mathbf{A}^g$, $\mathbf{S}^g$ for $g = 1, 2, \ldots, G$, $\mathbf{A}^*$, $\mathbf{E}$, $\mathcal{H}$, and $\mathcal{D}$.

Repeat

  for $g = 1, 2, \ldots, G$ **do**

    Calculate $\mathbf{W}_1^g$ with (10)

    Calculate W_2^g with N_g/P

    Update $\mathbf{A}^g$ with (15)

    Update $\mathcal{S}^g$ with (19)

  end

  Update $\mathbf{A}^*$ with (20)

  Update $\mathbf{E}$ with (21)

  Update $\mathcal{H}$ with (22)

  Update $\mathcal{D}$ with $\mathcal{D} \leftarrow \mathcal{D} + \mu(\mathcal{S} - \mathcal{H})$

  Update μ with $\mu = \min(\sigma\mu, \mu_{\max})$

until stopping criterion is met.

Output: The consensus endmember matrix $\mathbf{A}^*$ and abundance tensor $\mathcal{S}$.

C. Implementation Issues

The pseudocode of the proposed SANTF model for hyperspectral unmixing is described in Algorithm 1. To implement the SANTF method effectively, the following issues need to be discussed.

1) It is crucial to initialize the related variables, including $\mathbf{A}^g$, $\mathcal{S}^g$, $\mathbf{A}^*$, $\mathbf{E}$, $\mathcal{H}$, and $\mathcal{D}$. For the initial of $\mathbf{A}^g$, the popular strategy is vertex component analysis (VCA) [11]. $\mathbf{S}^g$ can be initialized by applying the fully constrained least-squares (FCLS) algorithm [46]. $\mathbf{A}^*$ is initialized by (15) based on the resulting $\mathbf{A}^g$, and $\mathcal{S}^g$ for $g = 1, 2, \ldots, G$. $\mathbf{E}$ is initialized by (21). Let $\mathcal{H}$ and $\mathcal{D}$ equal $\mathcal{S}$ and zero, respectively.
2) The stopping criterion contains twofold. On the one hand, the number of iterations should not exceed a preset maximum number ($T_{\max}$). On the other hand, the SANTF model is terminated once satisfying $\|\mathcal{S} - \mathcal{H}\|_\infty < \varepsilon$, where t is the number of iterations and ε denotes the error threshold.

D. Computational Complexity Analysis

In order to provide a more comprehensive description of the proposed algorithm, the subsection gives a detailed analysis of its computational complexity. Referring to Algorithm 1, the main computational burden is incurred during the updates defined by (15), (19), (20), (21), and (22), whose computational complexities are $\mathcal{O}(p_g BM + (B + p_g)M^2 + M^3)$, $\mathcal{O}(p_g BM + (B + p_g)M^2)$, $\mathcal{O}(GBM + BM^2 + M^3)$, $\mathcal{O}(BM^2 + M^3)$, and $\mathcal{O}(rcM^2 + \max(r, c)\min(r, c)^2 M)$, respectively, where p_g denotes the total number of the pixels in the gth superpixel. Thus, the computational complexity is $\mathcal{O}\left(t\left(\sum_{g=1}^G ((p_g + G)BM + (p_g + B)M^2 + M^3) + rcM^2 + \max(r, c)\min(r, c)^2 M\right)\right)$ after t iterations.

IV. Experimental Results and Discussion

Experiments are carried out using both synthetic and real HSI datasets to assess the effectiveness and significance of the SANTF method. To ensure a comprehensive evaluation, several comparative algorithms are utilized in the analysis. These include EIC-NTF [34], MV-NTF-TV [33], uDAS [20], TV-RSNMF [26], SGSNMF [44], MSNet [47], and MAC-U [48]. For the SANTF method, specific parameter settings are set as $T_{\max} = 500$ and $\varepsilon = 1e - 4$. Additionally, to enhance the reliability of the results, each experimental setup is executed ten times. Moreover, experiments except MSNet are conducted under the environment of MATLAB R2022a software and computer configuration Intel Core i7 CPU @ 2.10 GHz and 32.0-GB RAM. Besides, MSNet is performed under the environment of Python 3.8.10 and computer configuration NVIDIA GeForce RTX 4060 Ti GPU @ 8.0-GB RAM.

To effectively evaluate the performance of the various methods, two quantitative metrics are employed: the spectral angle distance (SAD) and the root-mean-square error (RMSE). Specifically, SAD and RMSE measure the accuracy of the estimated endmembers and the abundances, respectively, expressed as

$$\mathrm{SAD}_m = \arccos\left(\frac{\mathbf{A}_m^T \hat{\mathbf{A}}_m}{\|\mathbf{A}_m^T\| \, \|\hat{\mathbf{A}}_m\|} \right) \tag{23a}$$

$$\mathrm{RMSE}_m = \sqrt{\frac{1}{N} \left| \mathbf{S}_m - \hat{\mathbf{S}}_m \right|^2} \tag{23b}$$

where $\mathbf{A}_m$ is the mth reference spectrum, $\hat{\mathbf{A}}_m$ is the mth estimated endmember, $\mathbf{S}_m$ is the mth original abundances, and $\hat{\mathbf{S}}_m$ is the mth estimated abundances.

A. Synthetic Data Experiments

In this section, we utilize the spectral library provided by the United States Geological Survey (USGS) to create a synthetic dataset. The process for generating this synthetic data can be outlined in several steps.

1) Four spectral signatures are selected from the USGS library.
2) The simplified Hapke model [49] is used to produce a set of spectra, as illustrated in Fig. 2.
3) A scene is defined with dimensions of $z^2 \times z^2$ where $z = 8$, which is subsequently subdivided into smaller regions measuring $z \times z$.
4) Each of these regions is assigned one of the four endmembers, and each pixel is chosen randomly from the corresponding spectra set.
5) Mixed pixels are created through the application of a spatial low-pass filter measuring $(z + 1) \times (z + 1)$.

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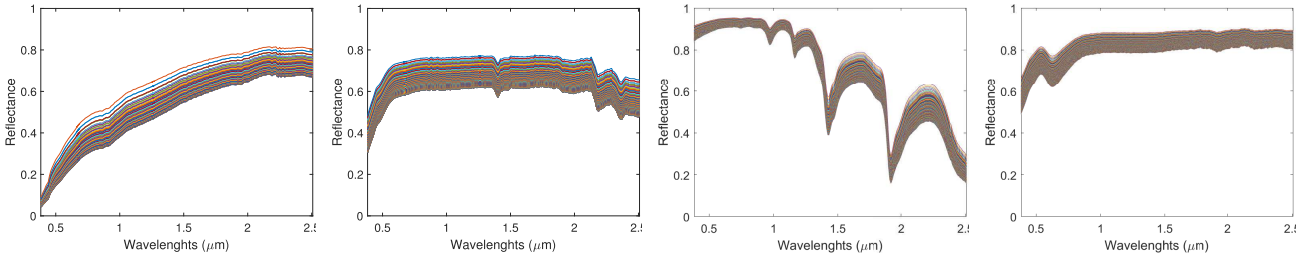


Fig. 2. Produced spectral signatures during our experiments. (a) Andradite WS488 (comprising 100 spectra). (b) Cookeite CAr-1.c < 30 μm (comprising 400 spectra). (c) Heulandite NMNH84534 (comprising 900 spectra). (d) Microcline NMNH135231 (comprising 1600 spectra).

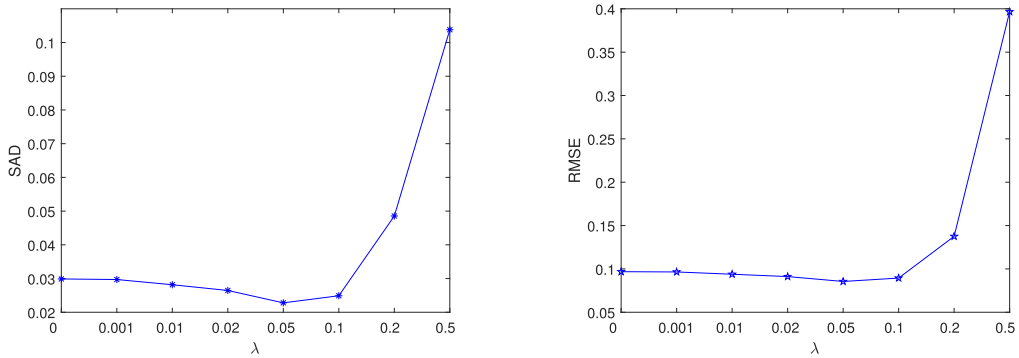


Fig. 3. Comparison of (a) SAD and (b) RMSE for our SANTF versus the parameter λ.

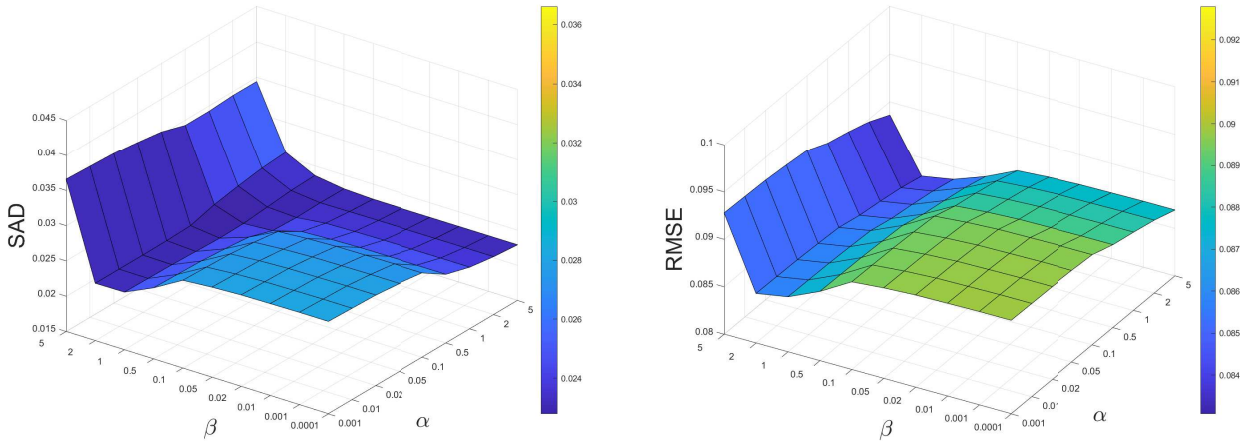
6) Pixels that exhibit abundances greater than a designated threshold value θ (defined here as 0.8) will be replaced by a mixture of all endmembers with equal weights.
7) The spectral data for each pixel are generated based on the LMM framework.
8) Finally, we introduce the Gaussian noise to the synthetic dataset to enhance its realism; the signal-to-noise ratio (SNR) is often employed to represent the different varying noise levels.

1) Experiment 1 (Parameter Analysis): To illustrate the effects of the parameters λ, α, and β, experiments are conducted using a synthetic dataset where the SNR is set to 30 dB. The value of λ varies within the range {0, 1e-3, 1e-2, 0.02, 0.05, 0.1, 0.2, 0.5}, while parameters α and β are, respectively, searched in given discrete values {1e-3, 1e-2, 0.02, 0.05, 0.1, 0.5, 1, 2, 5} and {1e-4, 1e-3, 1e-2, 0.02, 0.05, 0.1, 0.5, 1, 2, 5}. Fig. 3 presents the results obtained through SANTF in relation to the parameter λ, which controls the component of encoder. From it, we can observe the impact of the parameter λ on the SAD and RMSE. As λ increases from 0 to 0.1, the SAD and RMSE values remain relatively stable. Specifically, the SAD value is approximately 0.02 at $\lambda = 0.05$, while the RMSE value is about 0.1 at the same λ. However, when λ exceeds 0.1, both the SAD and RMSE values increase rapidly. At $\lambda = 0.5$, the SAD value reaches approximately 0.1, and the RMSE value reaches about 0.4. This indicates that the performance is improved gradually as λ increases, but deteriorates when λ is larger than 0.1. Therefore, λ can be set as 0.05 in this article. Besides, the SANTF performs better than that with $\lambda = 0$, which implies that integrating decoder is beneficial to enhance the unmixing performance. Fig. 4 illustrates the results for SANTF in terms of the parameters α and β. Obviously, an increase in β correlates with an upward trend in SAD and RMSE values. This indicates that parameter β significantly affects the model performance. Conversely, the effect of parameter α on SAD and RMSE appears to be relatively minor. Therefore, α and β can be set as 0.1 and 1, respectively.

2) Experiment 2 (Analysis of the Superpixel Size): To assess the effect of superpixel size on unmixing accuracy, we conduct experiments on a synthetic dataset corrupted by Gaussian noise with different SNRs and the Samson dataset. With $\lambda = 0.05$, $\alpha = 0.1$, $\beta = 1$, and $\omega_s = 0.3$, the average superpixel size ω was varied from 3 to 15. As illustrated in Fig. 5, the unmixing performance changes with ω. For SNR = 20 dB, both the SAD and RMSE values decrease as ω increases, whereas the results exhibit little sensitivity to ω for SNRs of 30 and 40 dB. In addition, it is evident that the Samson dataset yields higher SAD values than the synthetic dataset due to the complexity of real dataset. Obviously, both the SAD and RMSE values reach the optimal results when $\omega = 11$ for the Samson dataset. Consequently, $\omega = 7$ and $\omega = 11$ are determined for the synthetic and Samson datasets, respectively.

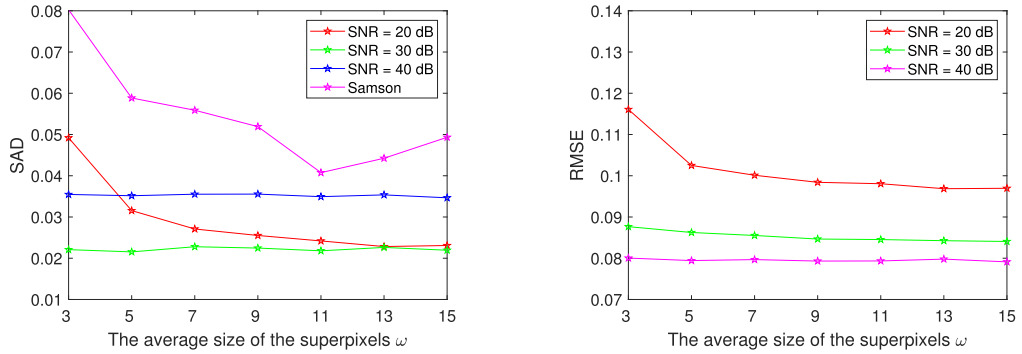
3) Experiment 3 (Robustness Analysis to Gaussian Noise): Here, we investigate synthetic datasets that incorporate varying degrees of Gaussian noise. In particular, hyperspectral data are affected by four distinct levels of Gaussian noise: 20, 30,

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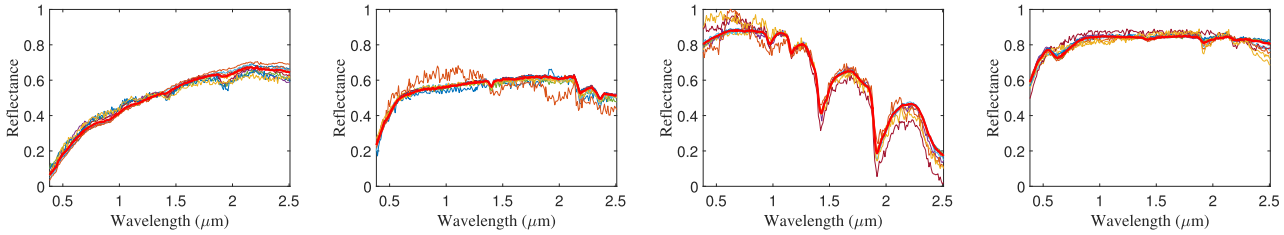
(a) (b)

Fig. 4. Comparison of (a) SAD and (b) RMSE for our SANTF versus the parameters α and β.



(a) (b)

Fig. 5. Comparison of (a) SAD and (b) RMSE for our SANTF versus the average size of the superpixels ω.



(a) (b) (c) (d)

Fig. 6. Randomly selected spectral signatures estimated by SANTF on the synthetic dataset with SNR = 40 dB. (a) Andradite WS488. (b) Cookeite CAr-1.c < 30 μm. (c) Heulandite NMNH84534. (d) Microcline NMNH135231. Red thick lines indicate the learned consistent endmembers.

and 40 dB. The corresponding SAD and RMSE values are presented in Table I. It can be observed that under different SNRs, various methods (SANTF, EIC-NTF, MV-NTF-TV, uDAS, TV-RSNMF, SGSNMF, MSNet, and MAC-U) exhibit different SAD and RMSE values. Generally, as the SNR increases, the SAD and RMSE values for all methods decrease, indicating that reduced noise levels contribute to the improved performance of these methods. In terms of SAD values, SANTF consistently outperforms the other methods across all three SNRs, particularly at 30 and 40 dB, where its values are significantly lower. Regarding RMSE values, both SANTF and uDAS demonstrate comparable performance at 40 dB; however, SANTF retains a distinct advantage at 20 and 30 dB. Overall, the SANTF method exhibits superior performance across various degrees of Gaussian noise, especially under high SNR conditions, suggesting that SANTF may be more effective in processing data with low noise levels. To illustrate the significance of considering spectral variability from the superpixel level, we present four spectral signatures estimated by SANTF on the synthetic dataset (SNR = 40 dB), as shown in Fig. 6. It can be seen that different endmember matrices can be generated from superpixels, leading to the formation

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TABLE I

SAD AND RMSE SCORES (AVERAGE OF TEN RUNS) ALONG WITH THEIR STANDARD DEVIATIONS ON THE SYNTHETIC DATASET (WITH DIFFERENT SNRS OF GAUSSIAN NOISE) FOR DIFFERENT METHODS. BOLDFACED NUMBERS DENOTE THE BEST RESULT UNDER EACH CONDITION

		SANTF	EIC-NTF	MV-NTF-TV	uDAS	TV-RSNMF	SGSNMF	MSNet	MAC-U
SAD	20 dB	**0.0411±2.20%**	0.0474±2.47%	0.0613±2.86%	0.0740±1.62%	0.0636±1.78%	0.0735±3.63%	0.0445±0.86%	0.0676±2.03%
	30 dB	**0.0252±0.37%**	0.0287±0.15%	0.0332±0.15%	0.0669±1.70%	0.0482±0.38%	0.0493±1.49%	0.0395±0.48%	0.0567±2.47%
	40 dB	**0.0316±0.15%**	0.0327±0.89%	0.0321±0.30%	0.0607±0.21%	0.0518±0.44%	0.0530±3.83%	0.0406±0.22%	0.0587±2.38%
RMSE	20 dB	**0.1135±2.39%**	0.1269±4.10%	0.1455±4.92%	0.1584±4.43%	0.1501±3.17%	0.1479±3.53%	0.1226±0.76%	0.1567±4.15%
	30 dB	**0.0907±0.95%**	0.0991±0.26%	0.1034±0.54%	0.1078±3.83%	0.1274±0.90%	0.1098±0.63%	0.1048±0.51%	0.0977±5.21%
	40 dB	**0.0968±0.61%**	0.1057±1.28%	0.1139±1.49%	0.0987±2.32%	0.1306±0.58%	0.1080±3.22%	0.1089±0.65%	0.0997±5.50%

TABLE II

COMPARISON OF RUNNING TIME (SECONDS) FOR CONSIDERED ALGORITHMS ON THE SYNTHETIC DATASET AND REAL DATASETS

	SANTF	EIC-NTF	MV-NTF-TV	uDAS	TV-RSNMF	SGSNMF	MSNet	MAC-U
synthetic dataset (SNR = 30 dB)	40.56	6.72	1.51	1.28	4.46	12.52	6.64	78.39
Samson	61.66	9.16	1.58	2.24	8.92	5.19	7.63	159.33
Cuprite	1217.94	172.88	577.74	85.41	106.48	45.71	21.35	4671.58
DLR HySU	0.38	0.79	0.17	0.31	0.87	1.18	5.96	29.35
Orchard	2.71	0.20	0.07	0.31	0.72	1.08	6.04	38.91

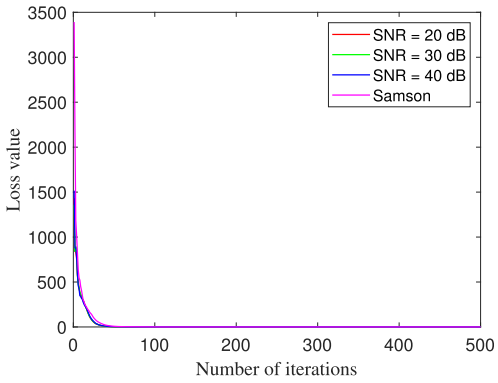


Fig. 7. Convergence curve of SANTF on the synthetic dataset and Samson dataset.

of endmember variability curves, as referenced by the thin curves in Fig. 6. Furthermore, by applying the designed DWE constraint, the consistent endmember spectra can be learned from these spectral variability curves.

4) Experiment 4 (Convergence Observation and Running Time): In order to validate the convergence of the proposed SANTF method, Fig. 7 displays the convergence curves on the synthetic dataset characterized by SNRs of 20, 30, and 40 dB and the Samson dataset. The curves clearly illustrate that the loss value, defined as $\|\mathcal{S} - \mathcal{H}\|_\infty$, decreases monotonically as the number of iterations increases. Additionally, these curves exhibit a rapid initial decline before both reaching a steady state around 50 iterations and then asymptotically approaching the preset maximum number of iterations ($T_{\max} = 500$). This consistent reduction in loss value confirms the convergence of the proposed SANTF method. In addition, we also investigate the average running time (seconds) of ten times for the various methods, shown in Table II, including the synthetic dataset with 30 dB, Samson, Cuprite, DLR HySU, and Orchard datasets. Notably, MSNet is implemented in Python and runs on a GPU, while the other methods are implemented in MATLAB and run on a CPU. Therefore, the time of MSNet is not comparable to that of the other methods. From Table II, all methods take the longest time on the Cuprite dataset because the Cuprite dataset has a large size and more endmembers. MAC-U exhibits the longest running times on all datasets, which may be due to its complex multilayer structure. The average running times of SANTF are significantly larger than those of EIC-NTF, MV-NTF-TV, uDAS, TV-RSNMF, and SGSNMF, but this is acceptable with the similar order of magnitude in time complexity. To improve the efficiency of the SANTF method in the future, we can focus on algorithm optimization, hardware acceleration, and parallel computing.

B. Real Data Experiments

1) AVIRIS Cuprite Dataset: This dataset is commonly applied to assess the efficacy of hyperspectral unmixing methods, shown in Fig. 8(a). It consists of 250 × 191 pixels, with each pixel representing 188 bands (specifically, 3–103, 114–147, and 168–220) following the removal of noisy bands. The range of wavelengths encompasses 0.4–2.5 μm. In this subscene, there are primarily 12 minerals identified: Alunite GDS82 Na82, Andradite WS487, Buddingtonite GDS85 D-206, Chalcedony CU91-6A, Kaolin/Smect H89-FR-5 30K, Kaolin/Smect KLF508 85%K, Kaolinite KGa-2, Montmorillonite + Illite CM37, Muscovite IL107, Nontronite NG-1.a, Pyrope WS474, and Sphene HS189.3B. The reference signatures utilized originate from the USGS spectral library.

Table III summarizes the results of this dataset for all methods. From Table III, it is evident that the various methods yield satisfactory mean values of SAD for the majority of materials. In addition, SANTF obtains the smallest results for five materials and mean SAD. This indicates that the overall

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TABLE III

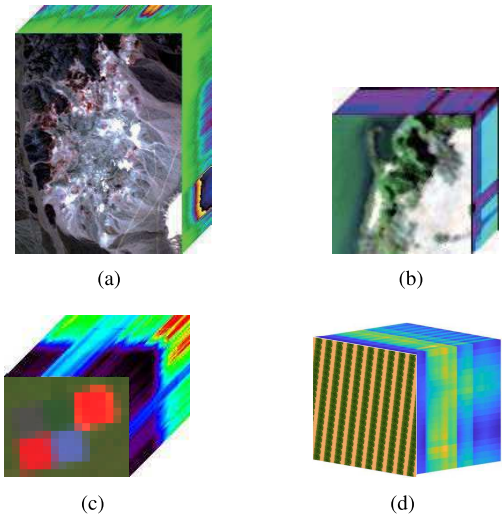
PERFORMANCE ABOUT SAD ALONG WITH STANDARD DEVIATION ON THE AVIRIS CUPRITE DATASET FOR DIFFERENT METHODS. BOLDFACED DENOTES THE BEST RESULTS UNDER EACH CONDITION

	SANTF	EIC-NTF	MV-NTF-TV	uDAS	TV-RSNMF	SGSNMF	MSNet	MAC-U
Alunite GDS82 Na82	0.1260±2.88%	0.1344±1.75%	0.1347±2.21%	0.1397±3.63%	0.1102±2.48%	0.1454±4.48%	0.1207±3.03%	**0.1016±1.88%**
Andradite WS487	0.0910±1.72%	**0.0730±1.30%**	0.0896±3.24%	0.0822±1.79%	0.0845±2.73%	0.0845±1.29%	0.0977±2.02%	0.0754±0.97%
Buddingtonite GDS85 D-206	0.0912±1.06%	0.0988±2.89%	0.0926±2.39%	0.0979±1.29%	0.0927±3.09%	**0.0864±0.87%**	0.1307±1.69%	0.0896±2.02%
Chalcedony CU91-6A	0.1395±1.92%	0.1418±1.63%	0.1469±1.39%	0.1449±0.69%	**0.1306±2.34%**	0.1337±2.23%	0.1635±3.72%	0.1543±1.62%
Kaolin/Smect H89-FR-5 30K	**0.0477±0.63%**	0.0944±2.48%	0.0881±2.24%	0.0529±1.83%	0.0775±2.32%	0.0837±2.65%	0.0839±3.01%	0.0778±2.10%
Kaolin/Smect KLF508 85%K	**0.0800±2.29%**	0.0925±2.50%	0.0982±4.63%	0.1267±2.03%	0.0973±3.28%	0.0830±0.78%	0.1349±3.90%	0.1016±3.18%
Kaolinite KGa-2	**0.1177±2.06%**	0.1748±5.24%	0.1541±5.89%	0.1406±2.35%	0.1607±4.87%	0.1453±3.80%	0.2273±4.93%	0.1614±3.83%
Montmorillonite + Illi CM37	**0.0540±0.30%**	0.0686±2.47%	0.0661±1.52%	0.0639±1.13%	0.0610±2.03%	0.0562±1.34%	0.0871±4.12%	0.0602±1.54%
Muscovite IL107	0.1001±0.57%	0.0987±1.35%	0.1013±0.69%	0.0941±0.60%	0.0958±0.84%	0.1070±1.37%	0.1218±3.82%	**0.0824±0.62%**
Nontronite NG-1.a	0.1144±0.39%	0.1026±3.63%	0.1369±1.53%	0.0994±1.68%	0.1142±1.98%	**0.0979±2.25%**	0.1281±3.23%	0.1180±3.06%
Pyrope WS474	**0.0884±1.38%**	0.0911±2.99%	0.0959±3.20%	0.0889±0.52%	0.1125±5.17%	0.1105±4.33%	0.1188±4.89%	0.1151±3.13%
Sphene HS189.3B	0.0859±2.06%	0.0651±0.92%	**0.0628±0.77%**	0.0880±2.73%	0.0770±2.46%	0.0663±1.13%	0.1290±9.34%	0.0785±2.81%
Mean	**0.0947±0.61%**	0.1030±0.64%	0.1056±0.78%	0.1016±0.47%	0.1012±0.76%	0.1000±0.72%	0.1286±1.13%	0.1013±0.61%

TABLE IV

SAD SCORES (AVERAGE OF TEN RUNS) ALONG WITH THEIR STANDARD DEVIATIONS ON THE SAMSON DATASET FOR DIFFERENT METHODS. BOLDFACED NUMBER DENOTES THE BEST RESULT UNDER EACH CONDITION

	SANTF	EIC-NTF	MV-NTF-TV	uDAS	TV-RSNMF	SGSNMF	MSNet	MAC-U
Soil	0.0092±0.07%	0.0295±0.00%	0.0305±0.18%	0.0196±0.01%	0.0311±0.10%	**0.0086±0.01%**	0.0438±2.02%	0.0465±0.06%
Tree	**0.0252±0.09%**	0.0550±0.05%	0.0538±0.19%	0.0470±0.15%	0.0524±0.10%	0.0412±0.21%	0.0326±1.26%	0.0280±0.01%
Water	0.0876±0.07%	0.1105±0.12%	0.1113±0.15%	0.1692±1.28%	0.1067±0.35%	0.1105±1.14%	**0.0486±0.82%**	0.0979±0.26%
Mean	**0.0407±0.03%**	0.0650±0.02%	0.0652±0.16%	0.0786±0.48%	0.0634±0.05%	0.0534±0.32%	0.0417±0.68%	0.0575±0.07%



(a) (b) (c) (d)

Fig. 8. Visualization of (a) Cuprite image, (b) Samson image, (c) DLR HySU image, and (d) Orchard image (shadow).

performance obtained by the SANTF method is better than that provided by other considered methods. Meanwhile, Figs. 9 and 10 show the endmembers and abundances estimated by SANTF method.

2) Samson Dataset: As shown in Fig. 8(b), the dataset consists of 95 × 95 pixels, with each pixel featuring 156 bands that range from 0.401 to 0.889 μm. There are three endmembers: soil, tree, and water.

Table IV lists the SAD values for all methods. As can be seen, the SAD values related to soil and trees extracted by all methods are considerably lower than those associated with water. This could be attributed to the more intricate signatures of water, which complicate proper initialization. Moreover, SANTF achieves superior spectra for tree in comparison to the other methods, yielding the smallest mean SAD. This demonstrates the effectiveness and advantage of the proposed method in addressing the unmixing issue. For illustration, Fig. 11 presents the plots of the reference signatures alongside the endmembers determined by the SANTF method on the Samson dataset. Fig. 12 gives the corresponding abundance maps.

3) DLR HySU Dataset: The DLR HySU dataset comprises an image of dimensions 86 × 123 pixels and 135 spectral bands, with wavelengths ranging from 0.417 to 0.903 μm, which can be downloaded by visiting the website: https://www.dlr.de/en/eoc/about-us/remote-sensing-technology-institute/photogrammetry-and-image-analysis/public-datasets/dlr-hysu. It includes six endmembers: Bitumen, Red Metal Sheets, Blue Fabric, Red Fabric, Green Fabric, and Grass [50]. In order to demonstrate the effectiveness of the proposed SANTF model, following the experiments in [51], a specific large target region of 13 × 16 pixels is selected, as shown in Fig. 8(c).

Table V facilitates a comparison of the SANTF method's performance against several considered methods, evaluating them using SAD scores (averaged over ten runs) and standard deviations on the DLR HySU dataset. Overall, all methods

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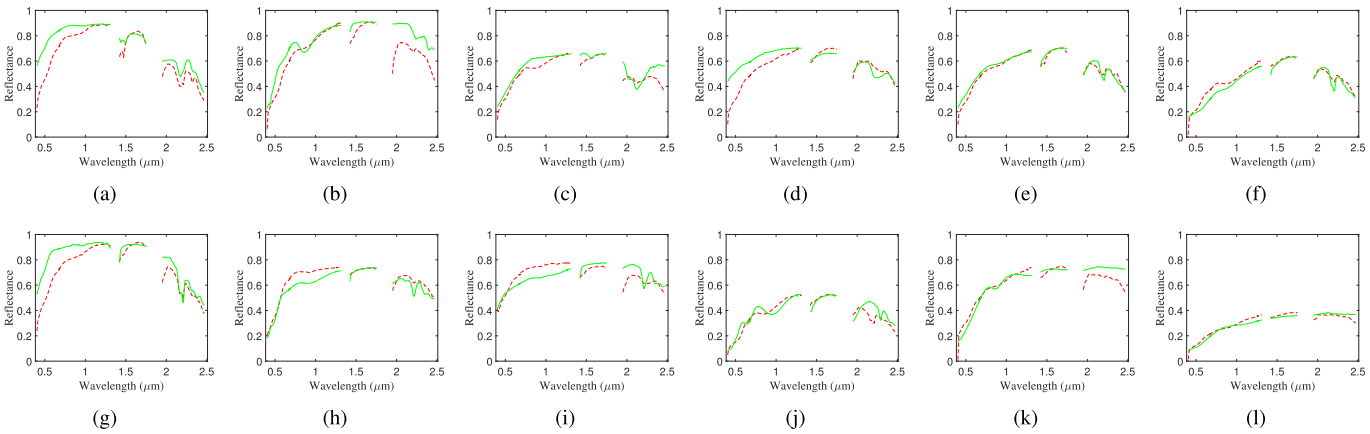


Fig. 9. Comparison of the USGS library spectra (green solid line) with those estimated by SANTF (red dashed line) on the AVIRIS Cuprite dataset. (a) Alunite GDS82 Na82. (b) Andradite WS487. (c) Buddingtonite GDS85 D-206. (d) Chalcedony CU91-6A. (e) Kaolin/Smect H89-FR-5 30K. (f) Kaolin/Smect KLF508 85%K. (g) Kaolinite KGa-2. (h) Montmorillonite + Illi CM37. (i) Muscovite IL107. (j) Nontronite NG-1.a. (k) Pyrope WS474. (l) Sphene HS189.3B.

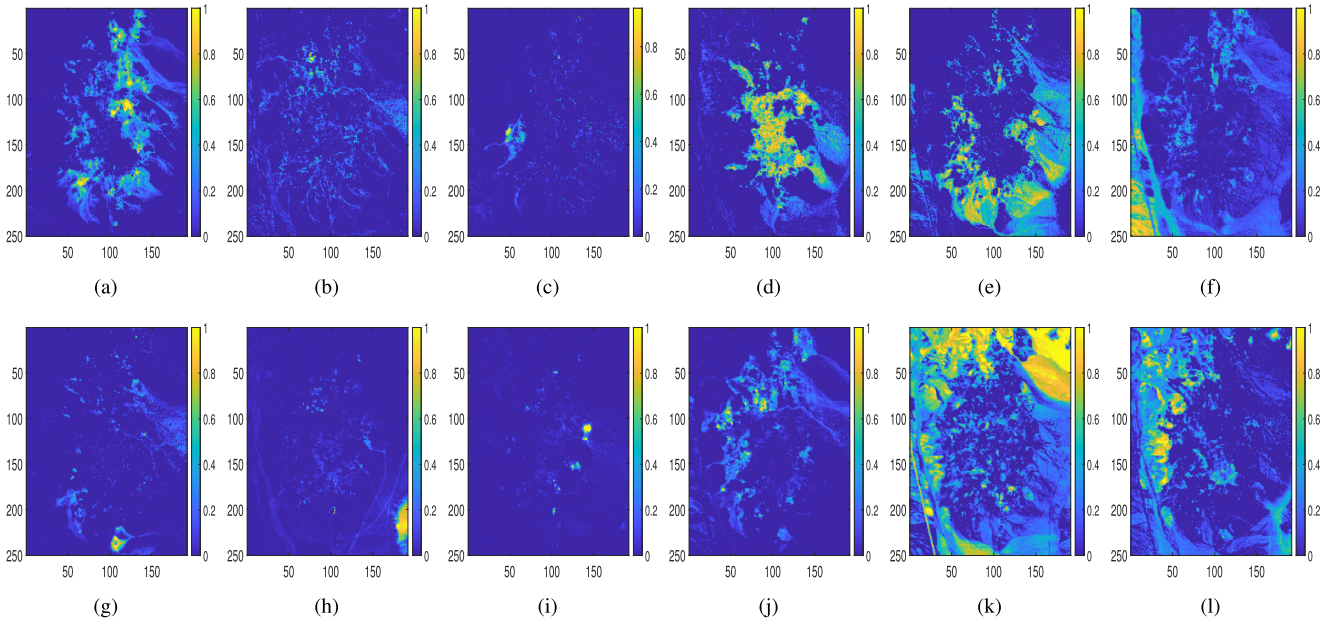


Fig. 10. Fractional abundance maps estimated by SANTF on the AVIRIS Cuprite subscene. (a) Alunite GDS82 Na82. (b) Andradite WS487. (c) Buddingtonite GDS85 D-206. (d) Chalcedony CU91-6A. (e) Kaolin/Smect H89-FR-5 30K. (f) Kaolin/Smect KLF508 85%K. (g) Kaolinite KGa-2. (h) Montmorillonite + Illi CM37. (i) Muscovite IL107. (j) Nontronite NG-1.a. (k) Pyrope WS474. (l) Sphene HS189.3B.

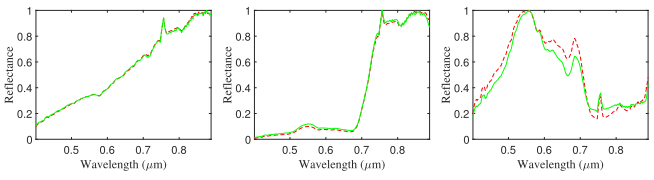


Fig. 11. Comparison of the reference spectra (green solid line) with those estimated by the SANTF method (red dashed line) of three endmembers on the Samson dataset. (a) Soil. (b) Tree. (c) Water.

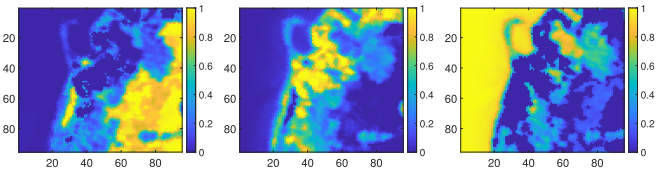
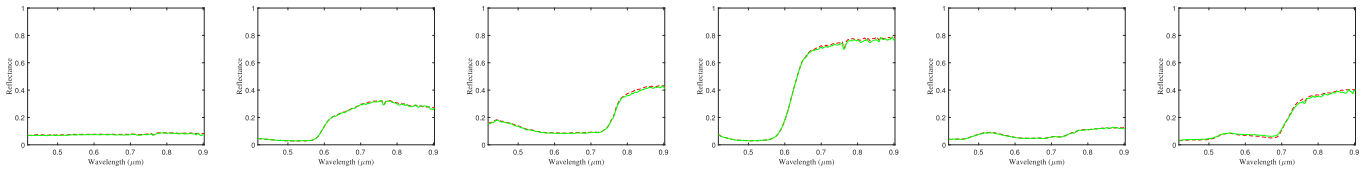


Fig. 12. Fractional abundance maps estimated by the SANTF method (red dashed line) of three endmembers on the Samson dataset. (a) Soil. (b) Tree. (c) Water.

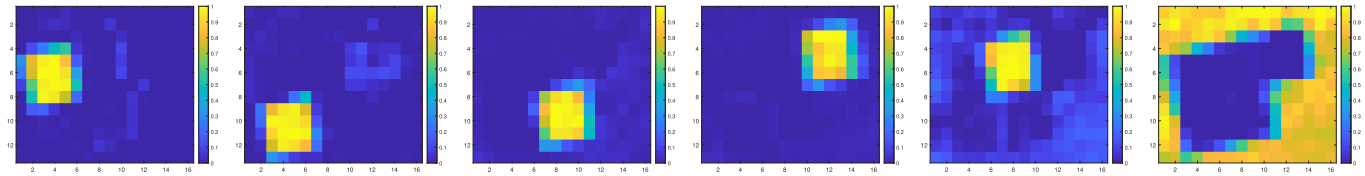
have achieved the satisfied unmixing performance, particularly for these three materials: Red Metal Sheets, Blue Fabric, and Red Fabric, possibly due to the simplicity of the scene and better initialization extraction. Nevertheless, SANTF obtains optimal SAD values for Blue Fabric and Red Fabric, and mean SAD value, indicating improved unmixing performance relative to other methods. For further insight, Fig. 13 displays reference signatures with the endmembers extracted by SANTF from the DLR HySU dataset, while Fig. 14 shows the corresponding abundance maps.

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(a) (b) (c) (d) (e) (f)

Fig. 13. Comparison of the HySpex library spectra (green solid line) with those estimated by SANTF (red dashed line) on the DLR HySU dataset. (a) Bitumen. (b) Red metal sheets. (c) Blue fabric. (d) Red fabric. (e) Green fabric. (f) Grass.



(a) (b) (c) (d) (e) (f)

Fig. 14. Fractional abundance maps estimated by SANTF on the DLR HySU dataset. (a) Bitumen. (b) Red metal sheets. (c) Blue fabric. (d) Red fabric. (e) Green fabric. (f) Grass.

TABLE V

SAD SCORES (AVERAGE OF TEN RUNS) ALONG WITH THEIR STANDARD DEVIATIONS ON THE DLR HYSU DATASET FOR DIFFERENT METHODS. BOLDFACED NUMBER DENOTES THE BEST RESULT UNDER EACH CONDITION

	SANTF	EIC-NTF	MV-NTF-TV	uDAS	TV-RSNMF	SGSNMF	MSNet	MAC-U
Bitumen	0.0245±0.05%	0.0363±0.45%	0.0289±0.13%	0.0693±0.11%	0.0297±0.07%	0.0714±0.47%	**0.0168±0.73%**	0.0319±0.05%
Red Metal Sheets	0.0150±0.00%	0.0153±0.05%	0.0160±0.03%	0.0493±0.02%	**0.0147±0.01%**	0.0190±0.10%	0.0219±0.74%	0.0152±0.01%
Blue Fabric	**0.0122±0.01%**	0.0139±0.10%	0.0129±0.07%	0.0346±0.08%	0.0157±0.04%	0.0144±0.01%	0.0213±0.57%	0.0179±1.71%
Red Fabric	**0.0081±0.05%**	0.0090±0.08%	0.0089±0.09%	0.0135±0.07%	0.0097±0.12%	0.0090±0.00%	0.0241±1.47%	0.0176±0.62%
Green Fabric	0.0233±0.01%	0.0275±0.16%	0.0265±0.10%	0.1106±0.03%	0.0308±0.12%	0.2102±0.18%	**0.0197±0.31%**	0.0324±0.02%
Grass	0.0364±0.37%	0.0346±0.31%	0.0335±0.34%	0.0298±0.41%	0.0240±0.24%	0.0403±0.53%	**0.0166±0.05%**	0.0380±0.50%
Mean	**0.0199±0.07%**	0.0228±0.11%	0.0211±0.05%	0.0512±0.08%	0.0208±0.06%	0.0607±0.00%	0.0201±0.26%	0.0255±0.37%

TABLE VI

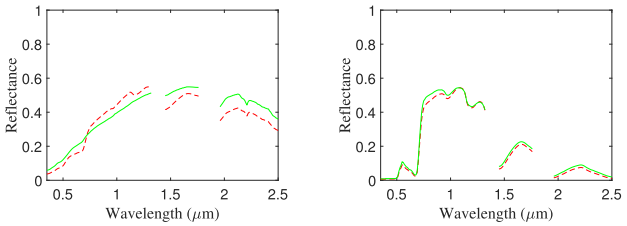
SAD AND RMSE SCORES (AVERAGE OF TEN RUNS) ALONG WITH THEIR STANDARD DEVIATIONS ON THE ORCHARD DATASET FOR DIFFERENT METHODS. BOLDFACED NUMBERS DENOTE THE BEST RESULT UNDER EACH CONDITION

		SANTF	EIC-NTF	MV-NTF-TV	uDAS	TV-RSNMF	SGSNMF	MSNet	MAC-U
SAD	Soil	0.1311±0.00%	0.1390±0.03%	0.1432±0.07%	0.1260±0.00%	0.1365±0.00%	**0.0631±0.00%**	0.1369±0.01%	0.1387±0.00%
	Tree	0.0420±0.00%	0.0399±0.07%	0.0359±0.00%	0.0607±0.00%	0.0480±0.00%	0.1122±0.00%	0.0382±0.01%	**0.0352±0.00%**
	Mean	**0.0866±0.00%**	0.0895±0.04%	0.0895±0.03%	0.0933±0.00%	0.0922±0.00%	0.0876±0.00%	0.0875±0.01%	0.0870±0.00%
RMSE	Soil	**0.0410±0.00%**	0.1157±0.03%	0.1456±0.10%	0.0511±0.00%	0.1253±0.00%	0.0778±0.00%	0.1223±0.04%	0.1186±0.00%
	Tree	**0.1125±0.00%**	0.2185±0.07%	0.2296±0.07%	0.1864±0.00%	0.2295±0.00%	0.2264±0.00%	0.2526±0.04%	0.2452±0.00%
	Mean	**0.0768±0.00%**	0.1671±0.03%	0.1876±0.08%	0.1187±0.00%	0.1774±0.00%	0.1521±0.00%	0.1874±0.03%	0.1819±0.00%

4) Orchard Dataset: As shown in Fig. 8(d), the Orchard dataset consists of 20 × 20 pixels, with each pixel featuring 185 bands [52], [53], [54], which can be downloaded by visiting the website: https://www.biw.kuleuven.be/nonlinear-mixing/scenes_orchard.html. The Orchard dataset contains two endmembers: soil and tree. Additionally, the Orchard dataset also provides the fractional abundances with shadow. Notably, the shadowed parts of the tree canopy are incorporated into the pure tree signature. In the experiments, the water vapor absorption bands (99–110 and 143–161) are excluded from the Orchard dataset.

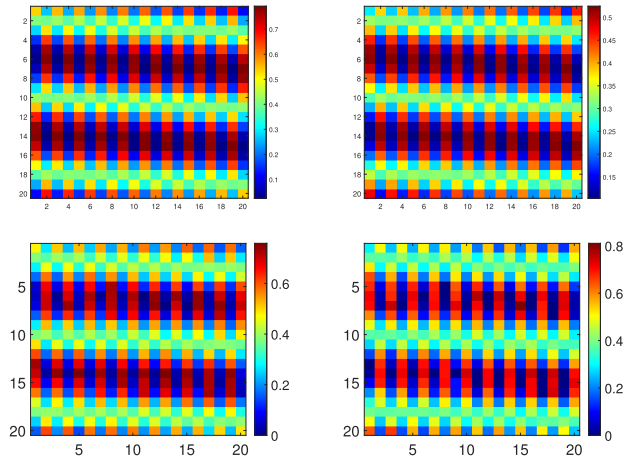
Table VI presents the SAD and RMSE values for all methods. As shown in Table VI, SANTF demonstrates superior unmixing performance compared to other methods in terms of both SAD and RMSE metrics on the Orchard datasets. Specifically, SANTF achieves the lowest mean SAD value, outperforming methods such as MSNet, MAC-U, and others. Additionally, since the Orchard dataset provides the true abundance maps, the RMSE metric is also calculated for our proposed SANTF method and the other compared methods. It can be observed that the mean RMSE value for SANTF is 0.0768 ± 0.00%, which is the lowest when compared to uDAS (0.1187 ± 0.00%), SGSNMF (0.1521 ± 0.00%),

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(a) (b)

Fig. 15. Comparison of the reference spectra (green solid line) with those estimated by the SANTF method (red dashed line) of two endmembers on the Orchard dataset. (a) Soil. (b) Tree.



(a) (b)

Fig. 16. Comparison of the reference abundance fractions (first row) with those estimated by the SANTF method (second row) of two endmembers on the Orchard dataset. (a) Soil. (b) Tree.

and other methods, further highlighting its better unmixing performance. This makes SANTF a highly effective method for the tasks evaluated in this study, demonstrating the significant effectiveness and advantage of the proposed method in addressing the unmixing issue. For illustration, Fig. 15 plots the reference signatures alongside the endmembers determined by the SANTF method on the Orchard dataset, while Fig. 16 gives the corresponding abundance maps.

V. Conclusion

In this article, a novel SANTF method is proposed for the hyperspectral unmixing task. To directly project the original data space into the abundance space, an ANTF model is constructed. To leverage local spatial information, superpixels are adopted into the ANTF model, and each superpixel cube generates one endmember matrix for better considering spectral variability. In this model, an rDWE is designed to adaptively learn the consensus endmember matrix. Furthermore, a TNN constraint is utilized to improve the low-rank characteristics of the abundance tensor. Comparing with the existing EIC-NTF, MV-NTF-TV, uDAS, TV-RSNMF, SGSNMF, MSNet, and MAC-U, the experimental results demonstrate that the proposed SANTF method outperforms the other considered methods. In the future, we will devote our efforts to exploring alternative superpixel generation methods to achieve more effective homogeneous regions, as well as constructing a general model to handle complex nonlinear mixtures.

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